- intellectual property rights, launch contracts, transfer of technology....

Hereafter, the 38th Colloquium on the Law of Outer Space was closed. The 39th Colloquium will be held during the International Astronautical Congress in Beijing, China, 7-12 October 1996.*

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IISL Secretary/Colloquium Coordinator

Comments

AT&T v. Martin Marietta: Further Reallocation of the Risk of Loss in Commercial Space Agreements

Introduction

On March 6, 1995, the world's largest telecommunications company, AT&T, sued the world's largest aerospace corporation, Martin Marietta, over the loss of AT&T's Telstar 402 telecommunications satellite. The complaint, filed in federal court in Alexandria, Virginia, alleges that Martin Marietta is guilty of fraud, bad faith, malice, knowing concealment of facts and the willful and wanton disregard of AT&T's rights, in addition to allegations of breach of warranty, breach of contract and negligent misrepresentation.[1] AT&T seeks over $400 million in compensatory damages and punitive damages of an unspecified amount. On the same day Martin Marietta sued AT&T in federal court in New Jersey[2] and in state court in Colorado seeking $8.9 million in compensatory damages for AT&T's alleged failure to pay Martin Marietta for the Telstar 402 spacecraft, and for a declaration by the court that Martin Marietta is not liable to AT&T for the Telstar 402 loss. Regardless of the outcome of this bitter battle between two giants, those portions of the AT&T-Martin Marietta contract for the Telstar 402 that appear in the pleadings filed in these actions show that there has been a further reallocation of the risk of

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* Information about the Colloquium, the session topics and the procedure for submission of papers, as well as the Manfred Lachs Space Law Moot Court Competition can be obtained from the IISL Secretariat, 3-5 rue Mario Nikis, 75015 Paris, France, tel. 33-1-45674260, fax 33-1-42732120.

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[1] AT&T Communications, Inc. v. Martin Marietta Corporation, C.A. No. 95-297A (E.D. Va.).

[2] Martin Marietta Corporation v. AT&T Communications, Inc., C.A. No. 95-989 (MLP) (D. N.J.).

loss in commercial space agreements in the decade and a half between 1980 and 1995.

A. *The Loss of Telstar 402*

AT&T, located in Bedminster, New Jersey, entered into a contract in August 1989 with General Electric for the design, construction, assembly, test and delivery of three telecommunications satellites, to be designated Telstar 401, 402 and 403. GE's Astro Space division at East Windsor, New Jersey, was to perform the spacecraft contract. In November 1990 AT&T entered into a launch services agreement with General Dynamics, Commercial Launch Services, Inc., located in San Diego, California. General Dynamics agreed to place two of AT&T's Telstar 4 satellites in geosynchronous orbit using Atlas launch vehicles. In 1993 Martin Marietta acquired GE's Astro Space Division. In December 1993 General Dynamics launched the first of AT&T's satellites, Telstar 401, and the satellite was successfully placed in geosynchronous orbit. In 1994 Martin Marietta acquired General Dynamics, Commercial Launch Services, Inc.

AT&T exercised an option in its launch services agreement and opted to launch Telstar 402 on an Ariane 42L launch vehicle. The launch took place at Ariane's facilities in Kourou, French Guiana on September 8, 1994. The launch vehicle performed normally and the spacecraft detached from it as planned. However, approximately 20 minutes after launch the spacecraft failed to attain its proper trajectory and was lost. The satellite was insured for $187.2 million at a premium of 18%. AT&T made a claim for a total loss on its satellite insurers and was paid the insured amount.

Martin Marietta established a Failure Review Board to look into the probable cause of the failure of Telstar 402. Because of several earlier satellite failures and some failures during ground testing that cast suspicion on the pyrovalves used in the fuel system of Martin Marietta's Series 7000 satellites, Martin Marietta engineers conducted more than 50 test firings and simulations of pyrovalves at its Engineering Propulsion Laboratory in Denver. In April 1995 Astro Space Vice President and 402 team leader Nathan J. Lindsay announced that Martin Marietta's tests established a "reasonable certainty" that the loss of Telstar 402 was caused by an explosion in the spacecraft's propulsion system that occurred when two redundant pyrovalves were fired sequentially to open a hydrazine fuel line shortly after the spacecraft's separation from the launch vehicle.[3] By-products from the explosive charge ignited hydrazine fuel that had accumulated on both sides of the second pyrovalve, resulting in an explosion that caused the spacecraft to fail completely, according to Lindsay. He said that Martin Marietta's tests confirmed that an excessive amount of combustible products, or "blow-by", is produced when the pyrovalve is fired, and that these gases can react with hydrazine to rupture the satellite's fuel line.

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[3] *See* "Pyrovalve Failure Blamed for Telstar 402 Loss," AV. WEEK & SPACE TECH., Apr. 17, 1995 at 24.

B. The Litigation

In the action filed in federal court in Alexandria, AT&T alleges, in addition to claims for the loss of Telstar 402, breaches of the Spacecraft Agreement and the Launch Agreement by Martin Marietta, and seeks damages for late delivery of Telstar 403 (renamed 402R) and for a decree of specific performance requiring Martin Marietta to make a launch slot available to AT&T for the future launch of Telstar 402R. This paper will focus only on AT&T's claims against Martin Marietta for the loss of Telstar 402, particularly its causes of action for breach of warranty and breach of contract.

Essentially, AT&T alleges that after the October 5, 1993 failure of Landsat 6, another Astro Space Series 7000 satellite using essentially the same propulsion system and nearly identical pyrovalves as Telstar 402, Martin Marietta failed to conduct a thorough failure review and did not perform a "fishbone" analysis of the cause of the Landsat 6 failure. According to AT&T, Martin Marietta advised it that the Landsat 6 failure was due to "adiabatic compression" and recommended as corrective action actuating the two redundant pyrovalves in the propulsion system of Telstar 402 30 seconds apart, rather than 1 second apart. However, according to AT&T Martin Marietta conducted tests, not disclosed to AT&T, which showed that "blowby", not adiabatic compression, caused the detonation of the hydrazine fuel. Furthermore, AT&T alleges, Martin Marietta engineers observed the blowby phenomenon on videotape while testing the same pyrovalve system for the Cassini program for NASA and the European space project. Despite this information, according to AT&T, Martin Marietta concealed this information from it and failed to conduct further tests and delay the production schedule in an effort to avoid overhead expense and negative publicity for its Astro Space division at East Windsor, New Jersey, gambling that the Landsat 6 failure would not recur in Telstar 402. These allegations are the basis of AT&T's claims for negligent misrepresentation, breach of the covenant of good faith and fair dealing, fraud and punitive damages. They also have relevance to AT&T's causes of action for breach of warranty and breach of contract.

In its cause of action for breach of warranty AT&T quotes Article 50.2 of the Spacecraft Agreement, which reads:

> In addition [to other remedies available under the Agreement], Contractor may be liable, up to a cumulative limit of two hundred million dollars ($200,000,000.00), for special, incidental, indirect or consequential damages, if, and only if, the following circumstances causally result in such provable damages to AT&T: (a) an officer or director of [Martin Marietta], the [Martin Marietta] Aerospace Group, or the [Martin Marietta] Astro Space Division or any of the Program Management Employees assigned to this Contract willfully or maliciously breaches material provisions of this Contract . . .

AT&T alleges that Martin Marietta is liable for damages for breach of warranty under Article 50.2 because it "willfully failed to correct the deficiencies in the design of the propulsion system that had caused the failure of Landsat 6. . . ."[4]

In its cause of action based on breach of contract, AT&T quotes Article 18.A of the Spacecraft Agreement, which provides:

> Without limiting the obligations of the Contractor under other provisions of this Contract, if the data resulting from in-plant system testing and integration, subcontractor testing and integration, launch vehicle integration, on-orbit acceptance tests or on-orbit operation shows that the spacecraft design does not meet all the performance requirements of this Contract, required to insure satisfactory and reliable operation of a spacecraft any time during the spacecraft's on-orbit design lifetime, the Contractor shall, promptly upon the request of AT&T, take appropriate corrective measures with respect to all the unlaunched spacecraft, whether or not previously delivered to or accepted by AT&T, so as to satisfactorily eliminate from the unlaunched spacecraft all design deficiencies noted in the launched spacecraft.

AT&T alleges that Martin Marietta breached the above Article by willfully failing to disclose critical information regarding the flightworthiness of Telstar 402 and the basis for its recommendation to change the firing sequence of the pyrovalves, which AT&T alleges actually exacerbated the danger of an explosion.

In its action filed in federal court in New Jersey, Martin Marietta alleges that any warranties it gave in the Spacecraft Agreement extended only "until Intentional Ignition" – defined as "the ignition of the first stage liquid engines of a Launch Vehicle that has been integrated with a Payload if said ignition is followed by the opening of the table clamps. . . ."[5] Martin Marietta alleges that the following disclaimer in the Spacecraft Agreement expressly excludes all other warranties:

> This warranty is in lieu of all other warranties, except as otherwise stated in this contract and except for a warranty of title, whether statutory, express or implied, including, but not limited to warranties of merchantability or fitness for a particular purpose.

Martin Marietta further alleges that, "The Telstar 402 failure was unintended and unanticipated. No Martin Marietta officer, director or

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[4] AT&T Complaint, para. 89, p.25.

[5] Martin Marietta's Complaint, para. 6, p.3.

employee willfully or maliciously breached any material provision of the [Spacecraft] Contract."[6]

Martin Marietta has filed a motion to dismiss the Alexandria action on the ground that AT&T failed to join an indispensable party – Martin Marietta Commercial Launch Services, a wholly-owned subsidiary of Martin Marietta.[7] In the alternative Martin Marietta has moved to transfer the causes of action concerning the Spacecraft Agreement to federal court in New Jersey and the causes of action concerning the launch agreement to federal court in Colorado. Failing that, Martin Marietta seeks to have the Alexandria action transferred in its entirety to federal court in New Jersey.

C. *Further Reallocation of the Risk of Loss in Commercial Space Agreements*

It is not the purpose of this brief paper to discuss the strengths and weaknesses of the positions of the two parties to this litigation, or to prognosticate about the ultimate outcome of these issues. Suffice it to note here that a comparison of the pleadings in *this AT&T v. Martin Marietta* litigation with two earlier cases shows that in the last decade and a half there has been a gradual but steady shifting of the allocation of the risk of loss in commercial space agreements away from the buyers to the manufacturer-sellers of commercial space products and the providers of launch services.

Appalachian Ins. Co. v. McDonnell Douglas Corp.[8] involved a 1980 contract between McDonnell Douglas, the manufacturer-seller of a third stage booster known as the PAM-D used in satellites launched from the Space Shuttle, and Western Union Corporation, the satellite owner and purchaser of the booster. At that time McDonnell Douglas had a monopoly on PAM-Ds, the only third stage booster that could be used when launching from NASA's Shuttle. Article 7 of the McDonnell Douglas-Western Union contract provided that, "...under no circumstances will [McDonnell Douglas] be liable to Purchaser [Western Union] under or in connection with this agreement, under any tort, negligence, strict liability, contract or other legal or equitable theory." After the failure of the PAM-D's exit cone resulted in a total loss of Western Union's satellite, Western Union collected its insurance and some of its satellite insurers filed a subrogation action against McDonnell Douglas based on strict liability in tort, negligence and breach of warranty, arguing that under California law strict liability in tort could not be contractually disclaimed. The appellate court in California ruled that the doctrine of strict liability in tort did not apply in the commercial setting of that case. It further found that the express warranty given by McDonnell Douglas's exit cone supplier,

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[6] *Id.* at 5.

[7] After the filing of AT&T's complaint on March 6, 1995, Martin Marietta became Lockheed Martin.

[8] 214 Cal. App. 3d 1, 262 Cal. Rptr. 716 (4th Dist. 1989).

Morton Thiokol, which warranty was to run to McDonnell Douglas "and to its customers and the users of its products," was abrogated by the above disclaimer because it had not formed a part of the "basis of the bargain" between McDonnell Douglas and Western Union. The Appalachian case, the first commercial space action brought against an aerospace manufacturer, represents the risk allocation pendulum in full swing in favor of the manufacturer-seller.

In *Martin Marietta v. INTELSAT*,[9] a case involving a mid-1980s launch services agreement between INTELSAT and Martin Marietta, the risk allocation pendulum slipped slightly away from the operator-buyer towards the manufacturer-seller. In that case INTELSAT's telecommunications satellite was left in a useless low earth orbit after its launch on March 14, 1990 when Martin Marietta's launch vehicle failed to separate the second stage from the payload because of faulty wiring by Martin Marietta's engineers. When INTELSAT demanded unlimited damages from Martin Marietta for its failure to perform under the launch services agreement, the manufacturer-seller filed a declaratory relief action based on the cross-waiver provisions in the Commercial Space Launch Act (CSLA).[10] In the launch services agreement Martin Marietta promised to make its "Best Efforts" to furnish launch services, which efforts were defined in the contract as "diligently working in a good and workman-like manner as a reasonable, prudent manufacturer of launch vehicles and provider of Launch Services." The Court of Appeals for the Fourth Circuit rejected Martin Marietta's argument that the replacement launch provision in the contract provided INTELSAT's sole and exclusive remedy, finding that the contract was ambiguous and remanding the case for a jury trial on the factual question of the parties' intention. The Court also found that the CSLA did not bar INTELSAT's claim based on gross negligence. The parties settled the dispute on remand to the trial court.

It can be seen that in the AT&T-Martin Marietta contract for the Telstar 402 spacecraft the risk of loss allocation pendulum has slipped further from the buyer and towards the manufacturer - seller. AT&T was able to obtain a clause in the contract making Martin Marietta liable if any "Program Management Employees assigned to this Contract willfully or maliciously breaches material provisions. . . ." Proof of such action is another matter, but the mere existence of the provision shows that commercial space buyers are beginning to extract more remedies from manufacturer-sellers than was previously the case. Similarly, the clause in the AT&T-Martin Marietta contract requiring the manufacturer to "take appropriate action with respect to all unlaunched spacecraft" if tests or in-orbit operation "shows that the spacecraft design does not meet all the performance requirements of this Contract," is a far cry from the language found in the McDonnell Douglas-Western Union contract for the PAM-D in 1980.

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[9] 991 F.2d 94 (4th Cir. 1993).

[10] 49 U.S.C. §§ 2601-2623 (1990).